

Applications of Statistical Modeling:

Module #1, Lecture #1

Course Overview; Multilevel Modeling Concepts

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Course Overview

- Welcome to SURV 617 / SURVMETH 687!
- We will begin today with a detailed overview of the **course syllabus**, available on the Canvas site for the course

Prerequisites

- Applied statistical methods
 - Univariate and multivariate distributions
 - Multiple regression
 - Logistic regression
 - Model diagnostics/interpretation
- Maximum likelihood estimation, (weighted) least squares estimation
- Hypothesis testing
- Applied sampling methods

Given those prerequisites, you should remember that...

- Linear regression: suppose the continuous outcome Y is measured for individual i ,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad \epsilon_i | X_i \sim_{iid} N(0, \sigma^2)$$

- Logistic regression: the outcome is a binary variable with values 0/1

$$\text{Log}\left(\frac{\Pr(Y_i=1)}{\Pr(Y_i=0)}\right) = \beta_0 + \beta_1 X_i$$

Recommended Readings

- *Regression and Other Stories*, Andrew Gelman, Jennifer Hill, Aki Vehtari, 2020, Cambridge press.
 - Accessible at <https://avehtari.github.io/ROS-Examples/>
- *Data Analysis Using Regression and Multilevel/Hierarchical Models*, Andrew Gelman and Jennifer Hill, 2006, Cambridge press.

Feel free to raise questions and comments about these readings at any point!

The ESS Data

The European Social Survey (ESS) Data

- Data were collected in face-to-face interviews from a national sample of 1,703 adults in Belgium
- Variables include respondent ID, interviewer ID, and 22 variables measuring attitudes and opinions of the respondents on various topics
- Also included are final respondent weights (based on a complex sample design), along with interviewer-specific response rates (percentage scale)
- Available in SAS, SPSS, Stata, and R formats on Canvas

Key Motivating Issues

- Account for interviewer effects:
 - Measure interviewer variation
 - Include interviewer-specific covariates
- Adjust for complex sample design features
- Incorporate survey weights
- ***Key point: observations are not independent and share some similar characteristics***
- Example of clustered data, where clusters are indicated by interviewer IDs

Multilevel Modeling

Multilevel Modeling

- **Multilevel models** are regression models that include **random effects** associated with either:
 - **Randomly Sampled Clusters** of units of analysis (in cross-sectional studies; e.g., census blocks)
 - **Randomly Sampled Subjects**, measured repeatedly (in longitudinal / repeated measures studies; e.g., volunteers for a clinical study)
 - **Both Clusters and Subjects** (in clustered longitudinal data sets; e.g., HRS)

What Are Random Effects?

- **Random effects** are random variables that allow the coefficients in a regression model to randomly vary depending on the cluster or subject under study
- For this reason, multilevel models are also known as:
 - Random coefficient models
 - Varying coefficient models
 - Subject-specific models
 - Hierarchical linear models (reflecting the data hierarchy of a clustered or longitudinal data set)
 - Mixed-effects models (reason will become apparent soon)

Random Effects, cont'd

- Because they are random variables, we need to define a probability distribution for the random effects → multilevel models have additional parameters to estimate!
- In many cases, clusters or subjects are not randomly sampled based on a probability sample design; we assume that they are a random subsample from a larger population (e.g., survey interviewers)

Example Model Specification

- Consider the following model for a continuous dependent variable Y , measured on person i within cluster j :

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + u_{0j} + u_{1j} x_{1ij} + e_{ij}$$

Fixed effects Random effects Error

Example Model Specification, cont'd

- The **fixed effects** are what you probably know from prior classes as *regression coefficients* or *regression parameters*
- These are *unknown constants* defining relationships between predictors and dependent variables that we wish to estimate
- The **random effects** are *random variables*; we therefore need to define their distributions!

Example Model Specification, cont'd

- The following are examples of commonly assumed distributions for the random effects and random error terms (see McCulloch and Neuhaus, 2011):

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + u_{0j} + u_{1j} x_{1ij} + e_{ij}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{01} & \sigma_1^2 \end{pmatrix} \equiv D \right)$$

$$e_{ij} \sim N(0, \sigma^2)$$

Errors independent of random effects (?)

Variance-covariance
Matrix of Random
Effects (D)

Multilevel (or Hierarchical) Specification

- An alternative way of specifying this model that some analysts prefer (used by the HLM and MLwiN software):

Level 1: $y_{ij} = \beta_{0j} + \beta_{1j}x_{1ij} + e_{ij}$

Random coefficients (**not parameters!**)

Level 2: $\beta_{0j} = \beta_0 + u_{0j}$

$$\beta_{1j} = \beta_1 + u_{1j}$$

When combined, we have the same model!

Why the Hierarchical Specification?

- This specification clearly defines the role of covariates measured at higher levels in multilevel models
- View each Level 2 equation for a random coefficient as an intercept-only regression model (where the DV is a random coefficient)!
- We try to **explain variance** in the random effects by adding the fixed effects of Level-2 covariates to these models!

Why the Hierarchical Specification?

- For example: $\beta_{0j} = \beta_0 + u_{0j}$
- Fit a model, and compute the estimated variance of the random intercepts: $\hat{\sigma}_0^2 = 2$
- Include a fixed effect of subject gender in the model (assume a longitudinal study):

$$\beta_{0j} = \beta_0 + \beta_2 MALE_j + u_{0j}$$

- Now, $\hat{\sigma}_0^2 = 1$; we explained 50% of the variance in the intercepts with gender!

Matrix Specification

- Matrix specification is important for understanding how models are estimated
- **Conditional** on a set of random effects, we have the following distribution for the **vector** of observations on subject (or cluster) j :

$$y_j | u_j \sim N(X_j \beta + Z_j u_j, R_j)$$

Design
matrix of
covariates

Fixed
effects
vector

(Smaller)
Design
matrix of
covariates

Random
effects
vector

Variance-
covariance
matrix of
random
errors

Implied Marginal Model

- **Marginally**, averaged across all random effects, we have the following distribution:

$$y_j \sim N(X_j\beta, Z_jDZ_j' + R_j \equiv V_j)$$

Vector of
expected means
for subject (or
cluster) j



Marginal variance-
covariance matrix of
observations on subject
(or cluster) j



- Note where the marginal variances and covariances of observed values come from!

Implied Marginal Model

- Important Observations:
 - A multilevel model **implies** the marginal model
 - When we include random effects in a model, the variances of the random effects in D need to be positive (i.e., the D matrix is **positive definite**)
 - We could also fit the implied marginal model **without** random effects, and estimate parameters describing the covariances of the observations (more flexible); no longer a **mixed model**, but allows for negative within-cluster correlations

Implied Marginal Model: Example

- In a model with random intercepts only and two observations per cluster, this is the implied marginal variance-covariance matrix :

$$V_j = \begin{pmatrix} \sigma_0^2 + \sigma^2 & \sigma_0^2 \\ \sigma_0^2 & \sigma_0^2 + \sigma^2 \end{pmatrix}$$

- Note that the covariance of the two observations is defined by the variance of the random intercepts! *It therefore can't be negative.* What if this is not the best model?

Implied Marginal Model: Example

- In these cases, the software will complain that you are trying to estimate a variance component which is negative, or set the component equal to zero
- As a solution, you can **fit the implied marginal model, rather than the multilevel model with random effects**, which in this case would have a *compound symmetry* structure for the random errors, allowing for negative covariances among the random errors
- We can no longer talk about the random effects in this case, however, or refer to this as a mixed model

Aliasing

- We also have to be careful about **aliasing** when specifying models, which leads to the case where parameters cannot be uniquely identified in estimation; don't over-specify the model!
- For example, we would not be able to include random intercepts **and** allow the random errors to have a compound symmetry structure (we are modeling the same covariance twice!!!)
- In this case, there would be infinitely many solutions for the parameter estimates, and the software will complain during estimation

Review!

- Multilevel model: mixed effects (fixed and random), conditional specification
- Marginal model: variance-covariance matrix
- Equivalent specifications:
 - Conditional: random intercept only
 - Marginal: covariance matrix with compound symmetry structure and positive correlation
- Parameters of interest (to be estimated): fixed effects, variances, and covariances

Model Building

- Generally two strategies used in practice:

1. Top-down Strategy

- Start with fully-specified model, including all fixed effects of interest
- Identify “optimal” variance-covariance structures for random effects and random errors
- Eliminate non-significant fixed effects
- Re-consider variance-covariance structures

Model Building, cont'd

2. Step-up Strategy (Variance Explanation)

- Start with an “empty” or “unconditional” model, with only a random intercept (and possibly random error)
- Add Level-1 predictors to explain variance in the random errors, and possibly allow their effects to randomly vary
- Add Level-2 predictors to explain variance in random effects at Level 2, possibly allowing their effects to vary at higher levels
- Add Level-3 predictors...

Modeling Practice

1. Model specification, clear notation!
2. Model estimation
 - Coefficients (or fixed effects) and their variances
 - covariance parameters
3. Model comparison
 - Hypothesis testing
 - Information criteria
4. Model diagnostics
5. Model interpretation: Inference/prediction

Estimation of Model Parameters

- Multilevel models are generally estimated using some form of **maximum likelihood estimation**. The basic idea:
 - Form the likelihood function, or the probability of the observed data, assuming independent subjects / clusters denoted by j :

$$L(\beta, \theta) = \prod_{j=1}^n f(y_j | X_j, \beta, \theta)$$

- f is the **probability density function** for a given dependent variable defined by the **marginal model** (e.g., multivariate normal), and θ = the variance-covariance parameters
- For some models (generalized linear mixed models), the likelihood first needs to be *approximated*

Estimation of Model Parameters

- Next, use iterative mathematical methods (e.g., Newton-Raphson) to find the parameter estimates that maximize the (approximate) likelihood. Two strategies:
 - **ML estimation.** Estimate the variance-covariance parameters first (define the fixed effects as a function of the covariates and the marginal variance-covariance matrix in the likelihood), and then use *generalized least squares (GLS)* to estimate the fixed effects:

$$\hat{\beta} = \left(\sum_j X_j' \hat{V}_j^{-1} X_j \right)^{-1} \sum_j X_j' \hat{V}_j^{-1} y_j$$

- **Restricted ML estimation (REML).** Same idea, but account for the loss of degrees of freedom from estimating the fixed effects first, then use GLS

REML vs. ML

- Distinction important for **normal** outcomes
- REML estimation is generally the default for normal outcomes, as it results in **unbiased estimates** of covariance parameters
- ML estimation is better for testing fixed effects (more on this later...), but results in **biased estimates** of covariance parameters
- Be careful to recognize the default of the software that you are using!

Variance Estimation

- Variance estimates for the estimated fixed effects can be computed using standard GLS results (biased, but why?):

$$\text{var}(\hat{\beta}) = \left(\sum_j X_j' \hat{V}_j^{-1} X_j \right)^{-1}$$

- Variance estimation for the estimated covariance parameters is not nearly as straightforward; can be based on the **Hessian matrix**, which is a by-product of the maximum likelihood estimation process; requires asymptotic (large-sample) assumptions
- Many authors now advocate Bayesian approaches for estimating the variance of covariance parameters (or for making inference more generally); see West and Elliott (2014)

Hypothesis Testing

- When fitting multilevel models, many hypothesis tests regarding the parameters are based on comparisons of *competing* models
- **Reference model:** A “full” or “saturated” model containing all parameters of interest; different from the null model where only intercepts are included
- **Nested model:** A “reduced” model, where some of the parameters in the reference model are constrained to zero

Likelihood Ratio Tests

- Compared to the reference model, does fitting the nested model substantially drop the likelihood (or make the observed data seem less likely)?
- Likelihood Ratio Tests: -2 **ML** log-likelihoods
 - Fixed Effects
 - Covariance Parameters

Likelihood Ratio Tests for **Fixed Effects**

- Simple idea, but assumes “large” sample of clusters and “large” samples within clusters
- **Null hypothesis:** selected fixed effects are all equal to zero (not important predictors)
- **Test statistic:** difference in final -2 **ML** log-likelihoods between nested and reference models (nested model has some fixed effects set to zero)
- Refer difference to chi-square distribution with q degrees of freedom, where q is difference between two models in number of fixed effects estimated

Likelihood Ratio Tests for Covariance Parameters

- **If null hypothesis does not specify that a covariance parameter is equal to the boundary of its parameter space (e.g., a variance is equal to zero):**
 - Same approach as for fixed effects, using REML
- **If testing that a variance is equal to zero (e.g., the variance of a given random effect):**
 - Compute test statistic, but refer to approximate **mixture** of chi-square distributions
 - Examples of this approach will be provided
 - Not widely implemented in software...

Approximate t-tests

- **Tests of null hypotheses that individual parameters are equal to zero:** divide estimate by estimated standard error, and refer statistic to a given t distribution or $N(0,1)$
- When using t distributions, degrees of freedom are only *approximate*, because we use an *estimate* of the marginal variance-covariance matrix when computing standard errors (failure to account for this uncertainty leads to bias!); many possible methods
- Not recommended for covariance parameters, but still produced by some software

Approximate F-tests

- Same basic idea as for t-tests, only extended to multiple parameters
- Same issues with approximate degrees of freedom (some software does not even compute these approximate tests!)
- Similar to likelihood ratio tests, useful for testing hypotheses about multiple parameters (e.g., fixed effects associated with a categorical predictor)

Other Testing Methods

- The exact null distribution of the likelihood ratio test statistic for **variance components** (in models with random intercepts only) has been derived by other researchers, and an R routine enables simulations from this null distribution (used by West and Olson, 2010)
 - Will be demonstrated in next lecture
- Bayesian approaches to inference are also possible (see West and Elliott, 2014)

Information Criteria (IC)

- IC are used to compare competing models that don't have a clear nesting relationship
- AIC, BIC, other adjusted versions; penalize for number of parameters being estimated
- “Smaller is better”: decision criterion
- No clear winner has been demonstrated in the literature to date; best to compare models based on multiple criteria

Model Diagnostics

- We need to check whether distributional assumptions for the model are satisfied before interpreting a “final” model
- Important diagnostic analyses to consider:
 - Examine standard (Q-Q, fitted-residual, etc.) plots of “conditional” studentized residuals, which account for predicted values of random effects (EBLUPs) and are scaled by the estimated SD of the residuals
 - Examine **influence diagnostics** for both individual observations and higher-level clusters (SAS farthest along, but not many software options at this point)
 - Examine Q-Q plots for predicted values of random effects (EBLUPs), to check for normality and outliers

Why Are Diagnostics Important?

- Tests of significance and other forms of inference based on probability models all rely on various distributional assumptions
- If these assumptions are violated, the methods used to make inferences may also be invalid (e.g., non-normal residuals → p-values should not be used to test hypotheses)
- We may see interesting patterns; consider the famous case of the BISON DATA in R (Canvas)!
 - <http://www4.stat.ncsu.edu/~stefansk/>

Predicted Random Effects: EBLUPs

- Empirical **B**est **L**inear **U**nbiased **P**redictors of the random effects in a model
- Keep in mind that we do not estimate random effects (they are random variables); we estimate their **variances** and **covariances**
- We can compute their predicted values based on our data: $\hat{u}_j = E(u_j | Y_j = y_j) = DZ_j' \hat{V}_j^{-1} (y_j - X_j \beta)$
- We can also compute a variance-covariance matrix for these predictions, for testing

Intraclass Correlation Coefficients

- ICCs are measures describing the homogeneity of responses within a cluster (or within a subject) that we can estimate
- **Two-level models, random intercepts only:**

$$ICC = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2}$$

- **Three-level models, random intercepts only:**

$$ICC_2 = \frac{\sigma_{0,2}^2 + \sigma_{0,3}^2}{\sigma_{0,2}^2 + \sigma_{0,3}^2 + \sigma^2}$$

$$ICC_3 = \frac{\sigma_{0,3}^2}{\sigma_{0,2}^2 + \sigma_{0,3}^2 + \sigma^2}$$

Missing Data

- While multilevel models are great for analyzing unbalanced data sets (due to missing data or unbalanced study designs), we do make assumptions about missing data
- Missing data are assumed to be **missing at random (MAR)**; that is, conditional on the other variables in a model, observations with missing data are only randomly different from observations without missing data
- Important to include as many relevant covariates for an outcome as possible to make this plausible

Centering Continuous Covariates

- Recall that the intercept in a model is the expected value of the outcome when all covariates are zero (this may not be realistic!)
- This interpretation is especially important when we fit models allowing the intercept to randomly vary across subjects/clusters
- It's generally a good idea to **center** continuous covariates, either at the overall mean or the cluster-specific mean of the covariate, to make the intercept more interpretable

Sample Size Considerations

- Simulations suggest that for cross-sectional studies, at least 30 clusters are needed with at least 30 observations per cluster (30/30) to have reliable parameter estimates in multilevel models
- Rules vary: 20/50, 50/20, 10/100, 100/10
- Generally close to 1,000 observations needed
- For longitudinal studies, better to use software / simulations; simulations are good practice in general!
 - www.umich.edu/~bwest/almmussp.html
- Having many clusters with only one observation can lead to estimation problems

Thinking Ahead...

- In the next class, we will focus on various software procedures that are capable of fitting multilevel models, and how to use them
- Please take a careful look at the assigned readings for next week
- Please think about possible team members and a data set for the final project!